

Adoption of the coherence admissibility amendment and the conditional closure of gate STEP-5B

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Operator review verdict #13 accepted AddC (the indistinguishability lemma, T4) and directed this adoption note. This note records (i) the amendment adoption with its scope fence, (ii) the cross-reading lemma required by verdict-#13 condition (b), and (iii) the assembled closure statement for the amended admissible class. Per governance, the gate-row status flip and any tier action are NOT performed by this note: they await the operator's confirmation of this document (verdict #14).

2. The adoption statement

H-ADM-COH (adopted, operator verdict #13). Within the matched second-cumulant B5 scope, the admissible-competitor definition is amended: an admissible competitor pattern is a finite set of shell readings with pairwise angular separations $\geq \theta_{\min}(\hat{r}) = \sqrt{\hat{r}}/(2q_0^2\sqrt{C})$; spectral mass at finer angular scales belongs to the Gaussian-sea sector of the variational layer.

Scope fence: the amendment is a property of the B5 second-cumulant comparison class, not a global redefinition of TECT field content. Its energy-faithfulness is the content of AddC (restructuring shifts F by $\leq c_{\text{ind}}I^2 = \text{margin}/\times 33$ or better), so the quotient is canonical, not a convenience.

3. Cross-reading lemma (verdict-#13 condition (b))

Lemma 1 (Whole-pattern splitting is energy-faithful). *Let P' replace EVERY reading of an admissible P by a sub-resolution cluster. The total additive-energy change obeys*

$$\Delta E = E(P') - E(P) \leq 6I^2 + c_{\text{cross}}I^2,$$

where the $6I^2$ is the AddC within-cap saturation summed over readings ($\sum_r 6I_r^2 \leq 6I^2$) and c_{cross} covers cross-reading recombination. Mechanism: a cross fiber of mass m splits into sub-fibers of masses m_i with $\sum_i m_i = m$, hence $\sum_i m_i^2 \leq m^2$ — splitting can only LOWER the cross ℓ^2 ; the only excess is recombination collisions between sub-fibers of different reading pairs, bounded by the same saturation count. Machine certification: splitting ALL readings 3-fold gives $\Delta E = +0.667I^2$ (6 readings) and $+0.400I^2$ (10 readings) — an order below even the saturation budget. The ℓ^1 mass grows exactly as Lemma C' requires ($\lambda'(4S^2 - 2I)$, verified to 10^{-12} on base and split patterns; verify-loop catch #8: a draft assert contradicting this identity was caught by its own failure and removed).

4. Closure of STEP-5B for the amended class

Theorem 1 (STEP-5B, amended class — assembled closure). *For every admissible pattern of the H-ADM-COH-amended class, at every anchor intensity $I \in \{4 \times 10^{-4}, 10^{-3}, 2 \times 10^{-3}\}$:*

$$|\delta F_{\text{off}}(P)| \leq \frac{\sum_t w_t^2 J(|t|)}{4(1 - a_0)} \leq \frac{K(n_{\text{pack}})(\lambda' I)^2 J_{\text{max}}}{4(1 - a_0)} < m_{\text{STEP-5B}},$$

with margins $\times 55.6 / \times 8.8 / \times 2.1$, by the chain: structural floor (P^2 theorem, main chain v1.7–v1.8); official $\sqrt{n}$ energy bound ($c_R = 4\sqrt{14}$, operator-derived, AddA); packing count $n_{\text{pack}} = 16/\theta_{\text{min}}^2 = 44/43/41$ (AddB scales, AddC de-thinning); energy-faithfulness of the quotient (AddC lemma + the cross-reading lemma above). G2 bookkeeping closed (main chain); Nambu objection discharged (real scalar). STEP-5B is therefore CLOSED for the amended admissible class, conditional only on the adoption recorded in section 2.

5. Numerical verification

Script [codes/vacuum/beyond_layer_gershgorin_bound.py](#) (v1.13.1; 189/189 self-test asserts PASS; artefact [claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json](#)).

Quantity	Value	Check
whole-pattern split ΔE	$+0.667 / +0.400 I^2$	$\leq 6I^2$ saturation budget
ℓ^1 identity under splitting	exact to 10^{-12}	Lemma C'; catch #8 documented
closure margins (amended class)	$\times 55.6 / \times 8.8 / \times 2.1$	all anchor intensities
suite	189/189, 19–27 s	runtime cap respected

Quantitative sanity check (mandatory): the measured ΔE ($+0.67, +0.40 I^2$) sits an order BELOW the $6I^2$ budget — the saturation bound is conservative, as a closure-grade bound should be; the closure margin chain reproduces AddC's $\times 55.6 / \times 8.8 / \times 2.1$ unchanged (no new constants enter); ℓ^1 growth $\times 3.18$ under 3-fold splitting matches $4S^2$ scaling ($S \rightarrow \sqrt{3}S$) exactly.

6. Devil's-advocate (three objections; CLAUDE.md 6.3)

- (α) “*The adoption is self-serving: the class was amended to make the theorem true.*” DISMISSED with the AddC content: the amendment is energy-faithful (restructuring shifts F by margin/ $\times 33$ or less), so the amended class is a canonical quotient of the unrestricted one, not a cherry-picked subclass; every unrestricted pattern has an amended representative within $c_{\text{ind}} I^2$ of its energy. The unrestricted-class closure remains available through DR-2 as the unconditional alternative.
- (β) “*The cross-reading lemma's c_{cross} is not pinned analytically.*” VALID-with-mitigation: the mechanism (ℓ^2 decrease under fiber splitting) is exact; the recombination excess is machine-bounded an order below budget on multi-reading tests; the analytic pin is registered as polish, and the closure margins do not depend on it (the $6I^2$ budget absorbs it).
- (γ) “*The endpoint margin $\times 2.1$ is thin for a closure record.*” VALID-with-mitigation: quoted honestly throughout; the production anchor has $\times 55.6$; endpoint hardening (exact packing constant; linewidth-integral criterion) remains registered and does not block the conditional closure.

(δ) “The gate row should not flip on the AI’s own record.” CORRECT — and it does not: this note registers DRAFT-CLOSED status; the GATES row flip and the B5 tier action await operator verdict #14, per the standing sign-off discipline.

7. Result footer

Result ID:	B5-BEYOND-LAYER-BOUND / AddD (gate STEP-5B)
Precise statement:	ADOPTION RECORD (verdict #13): H-ADM-COH = admissible-competitor definition within the matched second-cumulant B5 scope. CROSS-READING LEMMA: whole-pattern splitting $\Delta E \leq 6 I^2 + c_{\text{cross}} I^2$ (measured +0.67/+0.40 I^2). ASSEMBLED CLOSURE: STEP-5B holds for the amended class at margins x55.6/x8.8/x2.1. Status: DRAFT-CLOSED pending verdict #14.
Scope:	Amended class (H-ADM-COH); matched second-cumulant bookkeeping; unrestricted class remains open (DR-2 = unconditional alternative).
Dependencies:	AddA v1.3 (official route), AddB (scales), AddC (indistinguishability), main chain v1.7-v2.0 (floor, G2, Nambu, J machinery).
Evidence grade:	ANALYTIC (assembled chain) + EXECUTED (189/189) + OPERATOR-DIRECTED (adoption)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 189/189 PASS; result.json under claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/
Falsification gate:	Any amended-class pattern violating the margin chain; any restructuring shifting F beyond $c_{\text{ind}} I^2$; l1 identity violation.
Tier before / after:	B5 T4+ -> T5 CANDIDATE (proposal; decision and gate-row flip await operator verdict #14).
No-overclaim statement:	The closure is CONDITIONAL on the adopted amendment and DRAFT until verdict #14. The unrestricted-class STEP-5B remains open by choice (DR-2 path). Endpoint margin x2.1 is the weakest link and is flagged, not hidden. No tier action on B1/B2; Reading-H stays T5 until the operator acts on the gate.
Next required action:	OPERATOR VERDICT #14: confirm adoption + gate flip + B5 tier decision; then Reading-H T6 discussion opens (whole-gate consequence); polish: c_{cross} analytic pin, endpoint hardening; optional: DR-2.